Homework 2

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Name: YuanHao Tang Student ID: 3120265621

Some of the problems are collected from the examples in the book.

Problem 1

Let $\mathcal{A} \in \mathcal{L}(\mathcal{C}_2(t) \to \mathcal{C}_2(t))$. With respect to the bases $\alpha_1(t)$, $\alpha_2(t)$ and $\alpha_3(t)$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

the matrix representation is

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

Determine $\mathcal{N}(\mathcal{A})$ and $\mathcal{R}(\mathcal{A})$.

Solution

Solving $Ax = 0$ gives $x_2 = -2x_3$ and $x_1 = 3x_3$. The first two columns are independent, and the third equals -3 times the first plus 2 times the second. Thus

$$\boxed{\mathcal{N}(\mathcal{A}) = \operatorname{span}\{3\alpha_1(t) - 2\alpha_2(t) + \alpha_3(t)\}},$$

$$\boxed{\mathcal{R}(\mathcal{A}) = \operatorname{span}\{\alpha_1(t) + \alpha_3(t),\ \alpha_1(t) + \alpha_2(t) + 2\alpha_3(t)\}}.$$

Their dimensions are 1 and 2, respectively.

Problem 2

Show the following propositions:

(1) $A \in \mathbb{F}^{n\times n}$, then $S = \{x(t) : \dot{x}(t) = Ax(t)\}$ is a linear space and is isomorphic with $\mathbb{C}^n$, where

$$\dot{x}(t) \triangleq \left[\frac{dx_1(t)}{dt} \quad \frac{dx_2(t)}{dt} \quad \cdots \quad \frac{dx_n(t)}{dt}\right]^T.$$

(2) $A \in \mathbb{C}^{n\times n}$, $B \in \mathbb{C}^{m\times m}$ are given matrices, $\mathcal{T}_{A,B} : \mathbb{C}^{n\times m} \to \mathbb{C}^{n\times m}$ is defined as

$$\mathcal{T}_{A,B}(X) = AX + XB,$$

then $\mathcal{T} \in \mathcal{L}(\mathbb{C}^{n\times m} \to \mathbb{C}^{n\times m})$.

Solution

(1) Take $\mathbb{F} = \mathbb{C}$, consistently with the stated $\mathbb{C}^n$; for $\mathbb{F} = \mathbb{R}$, the corresponding space is $\mathbb{R}^n$. For solutions x, y on a common interval containing 0 and scalars a, b,

$$\frac{d}{dt}(ax + by) = aAx + bAy = A(ax + by).$$

The zero function also belongs to S, so S is a linear space. Existence and uniqueness for the constant-coefficient initial-value problem imply that the linear evaluation map $E : S \to \mathbb{C}^n$, $E(x) = x(0)$, is bijective. Its inverse is $c \mapsto x(t) = e^{At}c$. Hence $\boxed{S \cong \mathbb{C}^n}$.

(2) For any $X, Y \in \mathbb{C}^{n\times m}$ and $a, b \in \mathbb{C}$,

$$\begin{aligned}\mathcal{T}_{A,B}(aX + bY) &= A(aX + bY) + (aX + bY)B \\ &= a(AX + XB) + b(AY + YB) \\ &= a\mathcal{T}_{A,B}(X) + b\mathcal{T}_{A,B}(Y).\end{aligned}$$

Therefore $\mathcal{T}_{A,B}$ is linear.

Problem 3

Given the following two sets of bases for $\mathcal{V}^3$:

$$\begin{aligned}\epsilon_1 &= (1, 0, 1)^T, \quad \eta_1 = (1, 2, -1)^T, \\ \epsilon_2 &= (2, 1, 0)^T, \quad \eta_2 = (2, 2, -1)^T, \\ \epsilon_3 &= (1, 1, 1)^T, \quad \eta_3 = (2, -1, -1)^T.\end{aligned}$$

define the linear transformation as $\mathcal{A}\epsilon_i = \eta_i$, $i = 1, 2, 3$.

(1) Find the transition matrix from bases $\epsilon_1, \epsilon_2, \epsilon_3$ to bases η_1, η_2, η_3;

(2) Give the matrix representation for $\mathcal{A}$ with respect to the bases $\epsilon_1, \epsilon_2, \epsilon_3$;

(3) Give the matrix representation for $\mathcal{A}$ with respect to the bases η_1, η_2, η_3.

Solution

(1) Write $E = [\epsilon_1\ \epsilon_2\ \epsilon_3]$ and $H = [\eta_1\ \eta_2\ \eta_3]$. Using the transition convention $H = ET$,

$$\boxed{T = E^{-1}H = \frac{1}{2}\begin{bmatrix} -4 & -3 & 3 \\ 2 & 3 & 3 \\ 2 & 1 & -5 \end{bmatrix}}.$$

(2) Since $\mathcal{A}(E) = H = ET$, we have $\boxed{[\mathcal{A}]_\epsilon = T}$.

(3) The change-of-basis formula gives $\boxed{[\mathcal{A}]_\eta = T^{-1}[\mathcal{A}]_\epsilon T = T}$.

Problem 4

Denote by $\mathbb{R}_n[\lambda]$ the set of all polynomials with real coefficients and degrees not exceeding n. Let a linear transformation $\mathcal{A} \in \mathcal{L}(\mathbb{R}_n[\lambda] \mapsto \mathbb{R}_n[\lambda])$ be defined as

$$\mathcal{A}(f(\lambda)) = \frac{d}{d\lambda}f(\lambda), \qquad f(\lambda) \in \mathbb{R}_n[\lambda],$$

where $\mathcal{L}(\cdot)$ is the set of linear transformations. When $n = 2$, compute

(1) The matrix representation of $\mathcal{A}$ with respect to the basis $\alpha_1(\lambda) = 1+\lambda+\lambda^2$, $\alpha_2(\lambda) = 1+\lambda$, $\alpha_3(\lambda) = 1+2\lambda$.

(2) Determine $\mathcal{N}(\mathcal{A})$ and $\mathcal{R}(\mathcal{A})$ of $\mathcal{A}$.

Solution

(1) Since $1 = 2\alpha_2 - \alpha_3$,

$$\mathcal{A}\alpha_1 = \alpha_3, \quad \mathcal{A}\alpha_2 = 2\alpha_2 - \alpha_3, \quad \mathcal{A}\alpha_3 = 4\alpha_2 - 2\alpha_3.$$

Placing the coordinates of these images in columns gives

$$\boxed{[\mathcal{A}]_\alpha = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 4 \\ 1 & -1 & -2 \end{bmatrix}}.$$

(2) A polynomial has zero derivative exactly when it is constant, and all polynomials in $\mathbb{R}_1[\lambda]$ occur as derivatives. Therefore

$$\boxed{\mathcal{N}(\mathcal{A}) = \operatorname{span}\{1\}}, \qquad \boxed{\mathcal{R}(\mathcal{A}) = \operatorname{span}\{1, \lambda\} = \mathbb{R}_1[\lambda]}.$$

Their dimensions are 1 and 2, respectively.

Problem 5

Let $\mathcal{A}$ be a linear transformation from a four-dimensional linear space $\mathcal{V}$ to itself, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be a basis of the linear space $\mathcal{V}$, and the matrix representation of $\mathcal{A}$ with respect to this basis is

$$A = \begin{bmatrix} 5 & -2 & -4 & 3 \\ 3 & -1 & -3 & 2 \\ -3 & \frac{1}{2} & \frac{9}{2} & -\frac{5}{2} \\ -10 & 3 & 11 & -7 \end{bmatrix}.$$

(1) Find the matrix representation of $\mathcal{A}$ with respect to the following basis

$$\eta_1 = \alpha_1 + 2\alpha_2 + \alpha_3 + \alpha_4,$$
$$\eta_2 = 2\alpha_1 + 3\alpha_2 + \alpha_3,$$
$$\eta_3 = \alpha_3, \qquad \eta_4 = \alpha_4.$$

(2) Determine $\mathcal{N}(\mathcal{A})$ and $\mathcal{R}(\mathcal{A})$ of $\mathcal{A}$.

Solution

(1) From $[\eta_1\ \eta_2\ \eta_3\ \eta_4] = [\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4]P$,

$$P = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \qquad \boxed{[\mathcal{A}]_\eta = P^{-1}AP = \begin{bmatrix} 0 & 0 & 6 & -5 \\ 0 & 0 & -5 & 4 \\ 0 & 0 & \frac{7}{2} & -\frac{3}{2} \\ 0 & 0 & 5 & -2 \end{bmatrix}}.$$

(2) The first two columns vanish and the last two are independent, so both dimensions are 2, and

$$\boxed{\mathcal{N}(\mathcal{A}) = \operatorname{span}\{\eta_1, \eta_2\}}, \qquad \boxed{\mathcal{R}(\mathcal{A}) = \operatorname{span}\{r_1, r_2\}}.$$

Since $\eta_3 = \alpha_3$ and $\eta_4 = \alpha_4$, we may take

$$r_1 = \mathcal{A}\alpha_3 = -4\alpha_1 - 3\alpha_2 + \tfrac{9}{2}\alpha_3 + 11\alpha_4,$$
$$r_2 = \mathcal{A}\alpha_4 = 3\alpha_1 + 2\alpha_2 - \tfrac{5}{2}\alpha_3 - 7\alpha_4.$$